

Last time: $\mathbb{N}$ natural #

Other uses of $\mathbb{N}$:

- For indexing $a_1, a_2, \dots$
 $\longrightarrow (a_i)_{i \in \mathbb{N}}$
- For Summation $\sum_{i \in \mathbb{N}} a_i$ (Convention $\sum_{i \in \emptyset} a_i = 0$).

What if $B \neq \emptyset$?

$$\sum_{i \in A} a_i = \sum_{i \in A \cup \emptyset} a_i = \sum_{i \in A} a_i + \sum_{i \in \emptyset} a_i$$

why?
 $\hookrightarrow$ if $A \cap B = \emptyset, A, B \subseteq \mathbb{N}$.

$$\sum_{i \in A \cup B} a_i = \sum_{i \in A} a_i + \sum_{i \in B} a_i$$

For Products

$$a_1 a_2 \dots = \prod_{i \in \mathbb{N}} a_i$$

(Convention: $\prod_{i \in \emptyset} a_i = 1$).

Recursive definition

if $a_1 = 1$.

$$a_n = n \cdot a_{n-1} \quad (n \geq 2)$$

$$a_n = n!$$

$$\mathbb{Z} := \{\dots, -2, -1, 0, 1, 2, \dots\}$$

$$\mathbb{Q} := \{a/b : a, b \in \mathbb{Z}, b \neq 0, \gcd(a, b) = 1, b > 0\}$$

$$\mathbb{R} := \{n.a_1 a_2 a_3 a_4 \dots : a_i \in \{0, \dots, 9\}\}.$$

Thm. $\sqrt{2} \in \mathbb{R}$, but $\sqrt{2} \notin \mathbb{Q}$

Pf. Suppose a contradiction

that $\sqrt{2} \in \mathbb{Q}$ if $\sqrt{2} = \frac{a}{b}$. ($a, b \in \mathbb{Z}, b \neq 0$).

Multiply by b then square both sides.

$$2b^2 = a^2. \quad (*)$$

Then a^2 is an even number ($a^2 = 2 \cdot k$).

Then, if a were odd, a^2 is odd. Contradiction $\Rightarrow a$ is even.

By definition, $a = 2a_1$, $(*)$ becomes

$$2b^2 = 4a_1^2, \quad b^2 = 2a_1^2 \quad (**)$$

By the same argument, b is even $\Rightarrow b = 2b_1$.

then $(**)$ becomes $4b_1^2 = 2a_1^2 \Rightarrow 2b_1^2 = a_1^2$

By repeating, $a_1 = 2a_2 = 2(2a_3) \dots$

$$b_1 = 2b_2$$

Since every natural # only has finitely many prime # factors, this is a contradiction.



Since $\mathbb{R}$ and $\mathbb{Q}$ both satisfy P_1 - P_{12} , and since this theorem is true for $\mathbb{R}$, but false for $\mathbb{Q}$.

We must have used things beyond P_1 - P_{12}
(e.g. definition of $\mathbb{Q}$, even/odd, factorization, induction, etc.).

III. Functions

Def. A function takes an input some real #'s and assigns a single real # to each input.

The set of allowed inputs is the domain of the function.

Notation: $f: D \rightarrow \mathbb{R}$ ($D \subseteq \mathbb{R}$).

Ex.

$x \rightarrow \boxed{\text{mult. inv.}} \rightarrow f(x)$
function, $D: \mathbb{R} \setminus \{0\}$. ★ $\lim_{x \rightarrow 0} x^{-1} = \infty$

$x \rightarrow \boxed{\text{output of all prime \#}} \rightarrow f(x)$ &e. $f(6) = \{2, 3\}$.
Not a function. Not in $\mathbb{R}$.

$x \rightarrow \boxed{\text{Throw a die and get outputs}} \rightarrow f(x)$
Not a function the output is not unique.

Def 2 , A function is a collection of pairs of real #'s.

$$f: \{(a,b) : a,b \in \mathbb{R}\}$$

Such that if $(a,b), (a,c) \in f$, then $b=c$.

Systems of Functions

$$S_0 = \{f: D \longrightarrow \mathbb{R}\}$$

$$\text{If } f, g \in S$$

$$f+g(x) = f(x) + g(x)$$

$$f \cdot g(x) = f(x) \cdot g(x).$$